

1.2 Volumes Worksheet

1. Let R be the finite region bounded by the graphs of $y = \sqrt{x}$, $y = 8$, $x = 16$.

a) Determine the volume of the solid formed when R is revolved around the line $y = 8$.

We see that we have cross sections which are disks stacking up in the x -direction. Therefore, we have:

- b) Setup, but do not evaluate, a definite integral which represents the volume of the solid formed when R is revolved around the line $x = 16$.
- c) Setup, but do not evaluate, a definite integral which represents the volume of the solid formed when R is revolved around the line $x = 64$.
2. The graphs of $f(x) = 3e^x$, $w(x) = 12e^{-x}$, and the y -axis bound a region R in the xy -plane. Setup, but do not evaluate, a definite integral which represents the volume of solid obtained when R is revolved the line $y = 15$.

3. Let R be the finite region bounded by the graphs of $y = 2x$, $y = -3x + 60$, and $y = 0$. Set up, but do not evaluate, definite integrals which represent the volumes of the following solids.

a) The volume of the solid formed when R is revolved around the line $x = -5$.

b) The volume of the solid formed when R is rotated around the line $y = 24$.

c) The volume of the solid with base R for which the cross-sections perpendicular to the y -axis are squares.

4. Let R be the finite region bounded by the graphs of $y = \frac{100}{x^2}$, $y = 25$, and $x = 5$. Set up, but do not evaluate, definite integrals which represent the volumes of the following solids.

a) The volume of the solid formed when R is revolved around the line $x = 5$.

b) The volume of the solid formed when R is revolved around the line $x = 9$.

c) The volume of the solid with base R for which the cross-sections perpendicular to the x -axis are equilateral triangles. *Hint: If S represents the length of each side in an equilateral triangle then its area is equal to $\frac{\sqrt{3}}{4}S^2$ squares.*

5. Let R be the finite region bounded by the graphs $y = 12x^2$ and $y = 48x$. Set up, but do not evaluate, definite integrals which represent the volumes of the following solids.

a) The volume of the solid formed when R is revolved around the line $y = 220$.

b) The volume of the solid whose base is R for which cross-sections parallel to the x -axis are semicircles.

6. Let R be the finite region bounded by the functions $y = 4\ln(x)$, $y = 0$, $y = 2$, & $y = x + 1$. Set up, but do not evaluate, definite integrals which represent the volumes of the following solids.

a) The volume of the solid formed when R is revolved around the x -axis.

b) The volume of the solid formed when R is revolved around the line $x = -1$.